Hyunjong Choi – Real-Time System Design for Autonomous (and Intelligent) Systems

- Making systems predictable and safe when they interact with the real world

Real-Time Requirements

- Timing constraints imposed by the physical world
- D

Real-Time foundation for intelligent applications

D

Systems support for intelligent applications

- Research moves from Theoretical study to Applications
- CBS-based Linux Scheduling
- Analysis framework
- Effectiveness of weakly-hard
- The Autoware Foundation
- F1Tenth autonomous racing
- Smart IoT monitoring
    - Perception, localization, planning

Lab: Real-Time Embedded & Control Systems Laboratory

Joann Chen – Privacy & Anonymity Lab

Privacy Risks

D

Privacy Initiatives

D

Research Overview

- Differential Privacy (“tool”) used for many different systems
- D
- Privacy Risks & Machine Learning

Differential Privacy

- Goal is to make the probability almost the exact same without person in data, as if the person is in data
    - P(Joann has cancer)

Research Focus

- Bridging DP theory and its practical components
- Investigate privacy concerns in emerging machine learning models
    - Language models are neither secure or private
- Providence Graphs
    - Investigate security issues in AI models via system log

P

Bryan Donyanavard – Computer Systems Intelligence Lab (I'm here!)

- Systems for autonomy
- **Augmented reality**
- Cyberphysical applications
    - Safety, timeliness, efficiency
- Systems research
    - A lot of connected devices, not so many users, a network, low-end cellphones, smaller devices (i.e. smartwatches)
- Objectives: Robustness, responsiveness, efficiency

Challenge: Environmental Diversity

D

Challenge: Platform Complexity

- Computational complexity
- Algorithmic complexity
- Data-driven computer vision / machine learning types of software

Challenge: Network Variability

- Radio vs. WiFi vs. IoT Calls

Goal

- Balance tradeoff between performance and efficiency of all these different objectives

Research Methodology

D

Research Domains

1. Real Time environmental understanding
2. Operating Systems and Architecture for Autonomous Devices
3. Reservoir Computing in Edge Networks

Hoa T. Vu – Algorithms Research

Streaming Algorithms

- Problem: Data arrives one item at a time, too much to store
- Goal: Answer questions using far less memory than the data itself
- Example: Given a stream of IP addresses, how many distinct addresses have we seen?

- In modern computing, it's usually not possible to fit all the inputs in one single machine, with the especially large amount of data

Distinct Elements in a Stream

- New vs. Already Seen
- "How many YouTube videos"
- Diagram
    - "Each item is fed in, then discarded"
    - "Cannot store the set of addresses (too many to keep)"

Algorithms with Predictions

- Idea: Classical algorithms assume no help
    - Real-word data often has structure
    - What if we had a some prediction about the input (coming from an AI model)?
- Speed up Algorithms
- Improve performance

- Even if the prediction is bad, it is no worse than before
- If prediction is good, then it's much faster than before

Machine Learning Algorithms

- Design and analyze algorithms used inside ML systems (i.e. LLMs)
- Bridge rigorous algorithm design with practical ML

Support & Opportunities

Funded Research

What he's looking for

- Must enjoy math and algorithms
    - I like math (at least pure math), though not fully sure about algorithms
- Curious and like solving puzzles
    - I love jigsaw puzzles
    - I think I like puzzles in general. That might just be one of the main sources of appeal behind the math classes I took
- No prior research experience required

Yifan Zhang – Security and privacy in everyday software

- Recruiting PhD students

Hidden risks across software ecosystems

- Software supply chains
- Mobile systems
- Connected devices

- I believe he mentions "Virtual Machines" too (?)
- "Augmented analysis"

AI for finding subtle software bugs

- LineBreaker (ASE 2025)

- 123 newly discovered bugs
    - Across 154 Python and C 1000+ star repositories

- A wrong variable, function or operator can leave valid code with wrong behavior

- Small models screen code. A larger model examines context

Deceptive advertising inside mobile apps

D

Mainly recruiting PhD students

Interested in…

- Software security

- Privacy
- AI for program analysis

Salimeh Sekeh – Foundations of Reliable, Efficient, and Robust AI-driven Systems in Open World From Theory to Applications

- Open positions for MS and Undergraduate students

Physical AI and Open World Challenges

- Amazon delivery robot going around neighborhood and delivering packages

Lab Projects: Open-World Robotic

- M2OWL: Multimodal Open World Learning
- Cascading Unknown Detection Failures
- M2OWL for Urban Environments

Known Unknowns & Unknown Unknowns: AI's Blind Spot Problem

- Safety isn't a Feature, it's an Architecture
- The Devil is in the Downstream
    - "Downstream decisions"

Lab Projects: AI for Healthcare

- Rethinking Post-Surgical Wound Care with Multi-Agent Systems

Outreach and Educational Activities

YouTube: https://www.youtube.com/@SekehLab

Hajar Homayouni – Data Science Lab

Objectives

- Improving quality and reliability of medical data – essential for informed decision-making
- Enhancing interpretability of machine learning models to gain insights into decision-making processes

Data Science Lab

- Synthetic Medical Data Generation – EDA of medical research publications (?)
    - Privacy-Preserving Synthetic Medical Data Generation

        - ALM: Augmented Lagrangian Method
    - D
- Anomaly Detection and Explanation
- EDA of medical research publications
    - “Something chatbot”
    - “Citation-grounded answers”

Privacy-Preserving Synthetic Medical Data Generation

D

Anomaly Detection and Explanation

- Mutation operators
- A lot of research in this area in the medical domain

EDA of Medical Publications

D

Xin Zhang – Urban AI Lab

Enabling Human-Centered Urban Intelligence

Reserach Vision

- “How AI can understand urban dynamics…?”

1. Predict urban dynamics – model how cities evolve across space and time (“predict stuff like traffic, what will happen in the future”)
2. Understand human decisions – learn sequential behavior from observations
3. Structure decision knowledge – Identity reusable skills and concepts
4. Trustworthy decision-making –

Predicting Urban Dynamics

- Question: How can we model complex spatial-temporal dependencies to predict not only what will happen, but how urban systems may evolve?
- Approach:

Understanding Human Decisions

- Question: How can AI learn how people make decisions from observed behavior?
- Approach:

Structuring Decision Knowledge

- Question:
- Approach:

Trustworthy Decision Making

- Question:
- Approach:

Toward Trustworthy Urban Intelligence

- Understand Urban systems
- Learn Human Decisions
- Build Trustworthy AI

Jianshu Liu – Pisces Lab (Performant, Intelligent, Scalable Computing with Evolving Systems Lab)

Link to website: jianshuliu1721.github.io/projects/

Phase 1: System Development

- Architecting and deploying diverse cloud systems

Example: ADT Platform for Traffic Simulation

- https://github.com/PiscesLab

Phase 2: Empirical Investigation

- Employing timeline analysis to research performance anomalies

Phase 3: Intelligent Solutions

- Designing strategies to mitigate performance variations
    - Proactive Threads management
    - Intelligent Auto-scaling Framework
    - Some other stuff

Xiaobai Liu – Machine Vision and Perception Lab

Demo: Robot arm system to make fresh meals for you

- Ok, that would be so convenient for me
- Robots can also clean, etc.

Agent as Policy for Robotic Manipulation

- Robot can fold clothes and cloths **without any training data** (H O W)

Project 1/3: Embodied AI

- Vision – Building the foundation for robotic manipulation

Project 2/3: Whistle Extraction

- Deep Whistle
    - Patch-based residual architecture
- SAM-Whistle
    - Segmentation-driven extraction

Project 3/3: Homelessness Survey

- Detection & geospatial mapping
- Field verification

- How can we help homeless people find homes?

- Point of Time (POT)?

- Homeless people can kinda go all over the place

- How can AI help with this? Camera system all over cars

Wai Wang – Cyber Innovation Group / Wireless Technology Lab

Uberize your Wireless Access

- Time, something something
- (Oof, nothing he's saying is sticking)
- Game Theory, Auction Theory

Local Wireless Ecosystem: Uberize Token, Pinch Wireless Access

- D

Intelligent Wireless Networks Reserach From Bits to Semantics

- Bit: How are you?
- Semantic: h r u?
- **Implicit Neural Representation**

Intelligent Wireless Multimedia Networks: Model as Data

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Tao Xie – Computer Architexture and Systems Lab

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Shangping Ren –

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